

PROJECT OF AI FOR CYBERSECURITY

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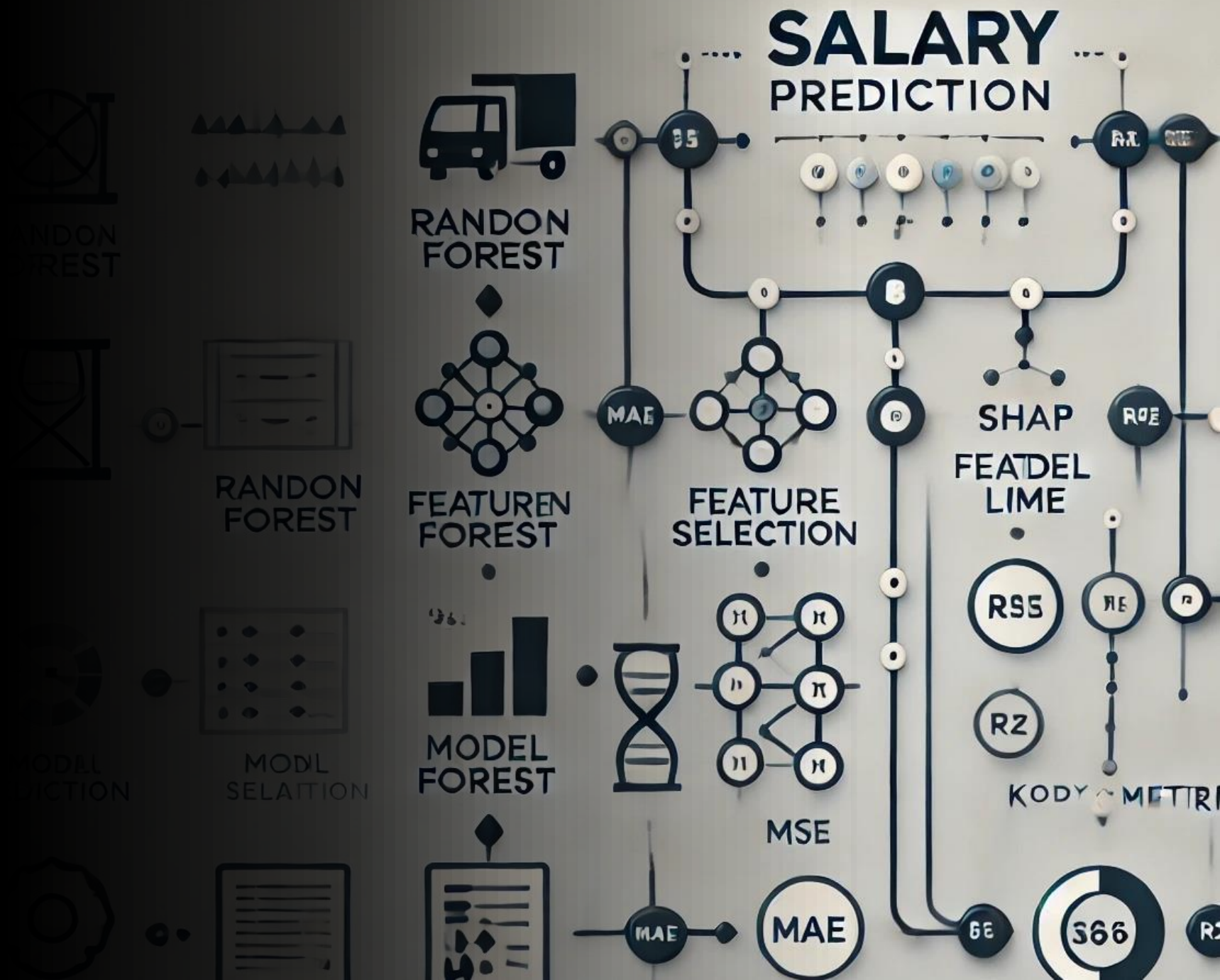


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Goal and specifics

Goal: Develop a machine learning model to **predict salaries** in the cybersecurity industry based on various features, enhancing accuracy and identifying key factors influencing salary variations.

Specifications:

1. Data Exploration

- Visualize numerical and categorical features
- Analyze relationships between features and salary
- Explore trends over time and regional disparities

2. Data Preprocessing

- Data cleaning:** remove duplicates, handle missing values
- Encoding:** OneHotEncoder and OrdinalEncoder
- Scaling:** Standard scaling for numerical features
- Outlier detection:** Apply log transformation to target

3. Model Development and Validation

- Models used:
 - Linear Regression**
 - Random Forest**
 - XGBoost**

4. Feature selection using:

- Recursive Feature Elimination (RFE)**
- Understanding Low Model Performance by Job Title**

5. Explainable Artificial Intelligence

- SHAP** (SHapley Additive Explanations)
- LIME** (Local Interpretable Model-agnostic Explanations)

Hyperparameter tuning with **RandomSearchCV**

Key Metric Goals:

Evaluate models using **K-Fold Cross Validation**

Focus on improving:

- Mean Absolute Error (**MAE**)

- Root Mean Squared Error (**RMSE**)

- R-Squared (**R²**)

Data Acquisition

Dataset

	salaries										
1	work_year	experience_level	employment_type	job_title	salary	salary_currency	salary_in_usd	employee_residence	remote_ratio	company_location	company_size
2	2024	MI	FT	Security Consultant	211000	USD	211000	US	0	US	M
3	2024	MI	FT	Security Consultant	142000	USD	142000	US	0	US	M
4	2024	MI	FT	Security Consultant	64417	GBP	80521	GB	0	GB	M
5	2024	MI	FT	Security Consultant	52584	GBP	65730	GB	0	GB	M
6	2024	MI	FT	Consultant	188400	USD	188400	US	0	US	M
7	2024	MI	FT	Consultant	125600	USD	125600	US	0	US	M
8	2024	MI	FT	Manager	246400	USD	246400	US	0	US	M
9	2024	MI	FT	Manager	117300	USD	117300	US	0	US	M
10	2024	MI	FT	Security Engineer	200200	USD	200200	US	0	US	M
...											
22600	2020	SE	FT	Information Security Officer	35000	USD	35000	AR	100	AR	L
22601	2021	EN	FT	Application Security Engineer	65000	USD	65000	US	100	US	L
22602	2020	SE	FT	Information Security Specialist	170000	USD	170000	US	100	US	L
22603	2021	SE	FT	Application Security Engineer	135000	USD	135000	US	100	US	L
22604	2021	EN	FT	Cyber Security Analyst	100000	USD	100000	US	50	US	M
22605	2020	MI	FT	Ethical Hacker	356000	GBP	456621	GB	100	GB	L
22606	2020	MI	FT	Cyber Security Analyst	140000	AUD	96422	AU	50	AU	M
22607	2021	SE	FT	Information Security Manager	60000	GBP	82528	GB	50	GB	L
22608	2021	SE	FT	Penetration Testing Engineer	126000	USD	126000	US	100	US	L
22609	2021	MI	FT	Information Security Analyst	42000	GBP	57769	GB	100	GB	L
22610	2021	MI	FT	Threat Intelligence Analyst	66310	USD	66310	US	0	US	L

Platform: Kaggle

Reason for Choosing This Dataset:

The dataset provides insights into the evolving landscape of cybersecurity roles, highlighting salary disparities based on experience, location, and job titles.

Dataset Description:

The dataset contains over **20,000 records** from various countries, with a strong prevalence of data from the United States.

It spans from **2020 to 2024**, making it highly recent and relevant to current industry trends.

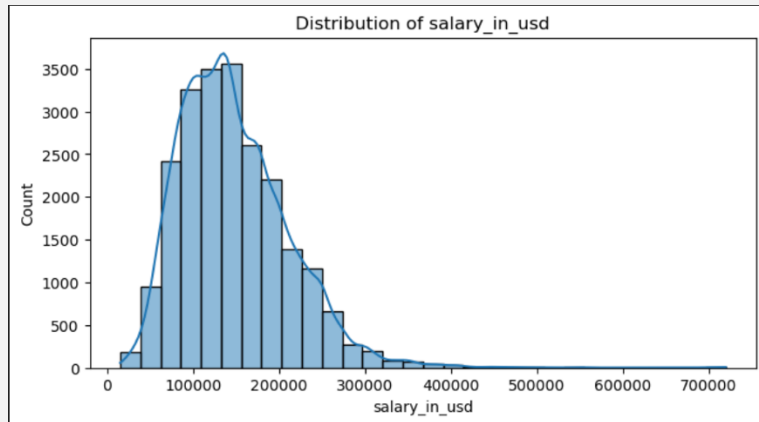
The data includes both **continuous** and **categorical variables**.

Data Exploration

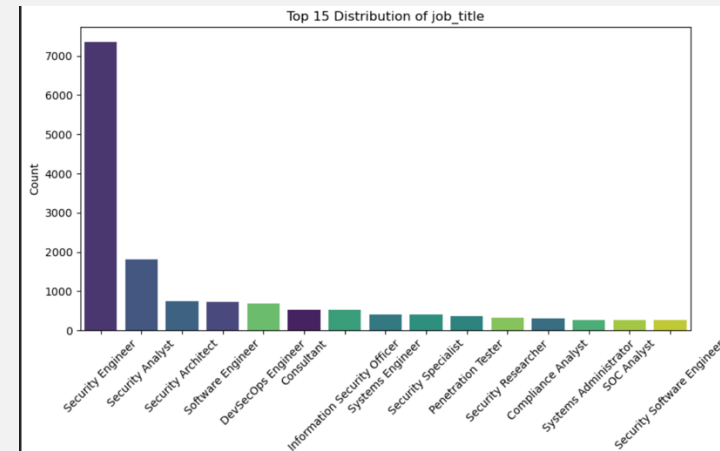
Objective:

Understand the dataset through visual and statistical analysis to identify patterns, correlations, and key insights that may influence salary predictions.

1. Visualizing Numerical Features



2. Visualizing Categorical Features



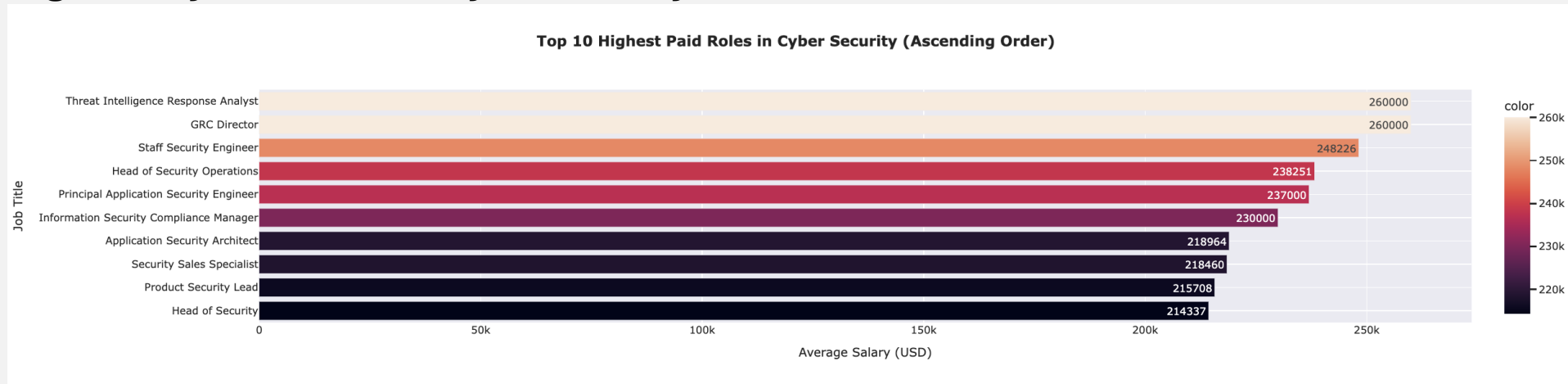
3. Relationship Between Categorical Features and Salary

4. Relationship Between Numerical Features and Salary

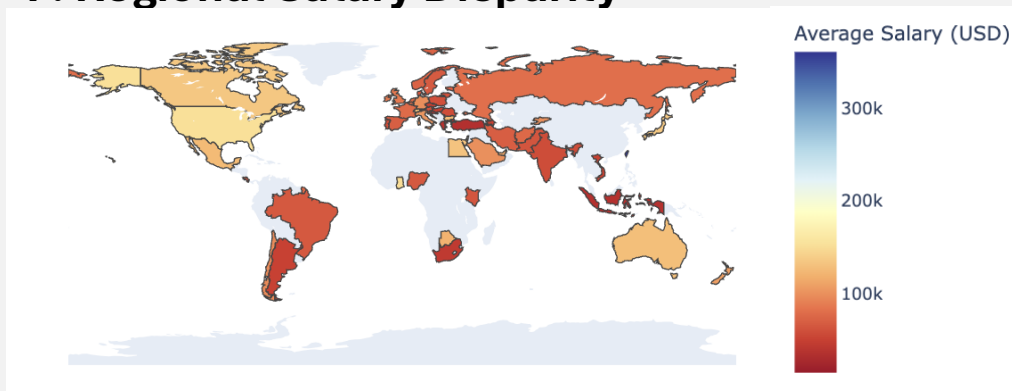
5. Salary Trends Over Time

Data Exploration

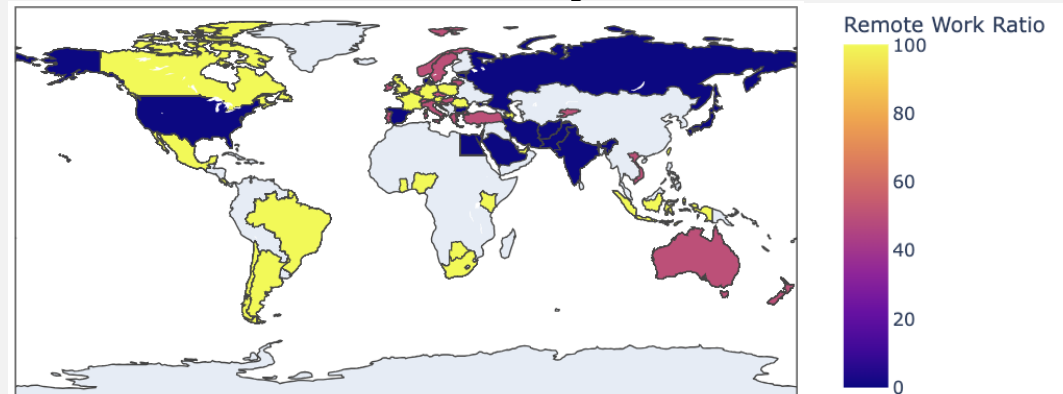
6. Average Salary for Different Cybersecurity Roles



7. Regional Salary Disparity



8. Remote work and salary



Data Preprocessing – Data Cleaning

Objective:

Ensure data quality by addressing duplicates, missing values, and high-cardinality features to improve model performance.

Duplicates Removal

```
Checking for Duplicates...  
Number of duplicate rows: 9333  
Duplicate rows removed.
```

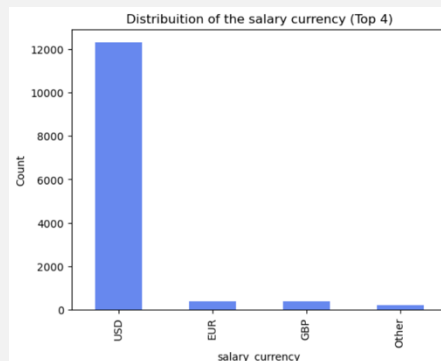
Identified and removed duplicate records to ensure unique observations.

Handling Missing Values:

```
Final Check for Missing Values:  
work_year      0  
experience_level 0  
employment_type 0  
job_title       0  
salary          0  
salary_currency 0  
salary_in_usd   0  
employee_residence 0  
remote_ratio    0  
company_location 0  
company_size    0  
dtype: int64
```

Missing values were imputed or dropped depending on their relevance to the analysis.

Reducing High Cardinality in Categorical Features:



1. For categorical columns **with numerous unique values** (e.g., employee_residence), only the top 4 most frequent categories were retained.
2. All remaining categories were grouped under "Other" to simplify the dataset and prevent overfitting.

Data Preprocessing - Encoding

Objective:

Convert categorical features into a numerical format suitable for machine learning models.

1.OneHotEncoder – Applied to features with a small number of unique values.

1. Each unique category is represented as a binary column (0 or 1).
2. Prevents losing valuable information from low-cardinality features without introducing bias.

2.OrdinalEncoder – Used for features with a logical order.

1. This method assigns an integer to each category, preserving its rank or significance.
2. helps manage high-cardinality features by avoiding an explosion of dimensions, keeping the dataset compact and manageable.

	employment_type_FT	employment_type_Other
0	1.0	0.0
1	1.0	0.0
2	1.0	0.0
3	1.0	0.0
4	1.0	0.0

	experience_level	job_title
0	3.0	188.0
1	3.0	188.0
2	3.0	140.0
3	3.0	104.0
4	2.0	178.0

Data Preprocessing – Outlier detection and scaling

Objective:

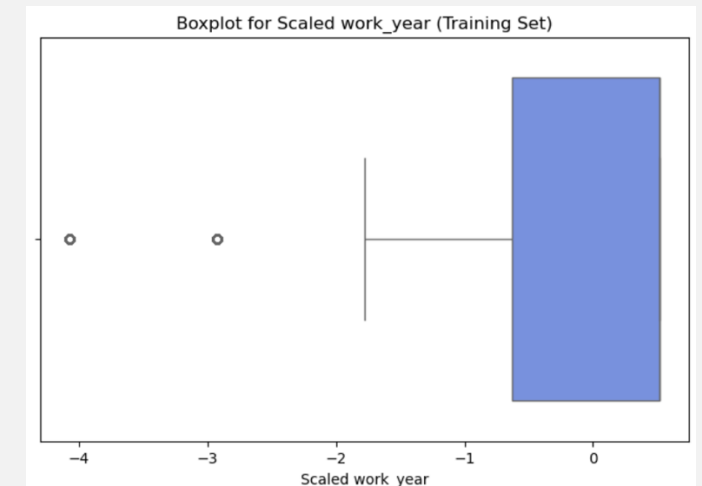
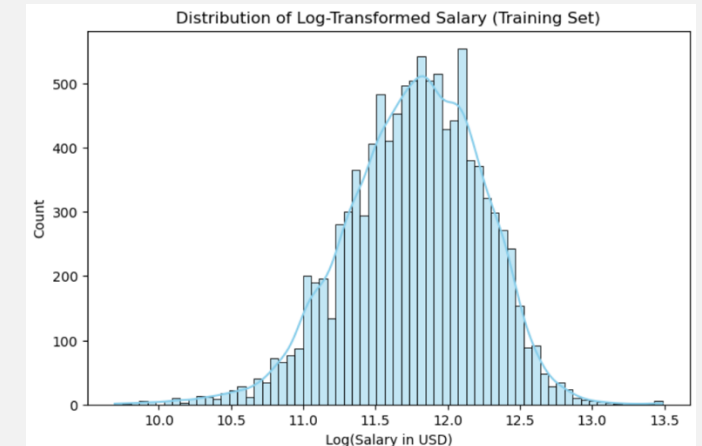
Normalize the data to ensure models are not biased by extreme values and to bring numerical features to the same scale.

1.Outlier Detection (Log Transformation):

1. A log transformation was applied **only to the target variable** (salary_in_usd).
2. This helps reduce the skewness caused by extreme salary values, stabilizing model predictions.

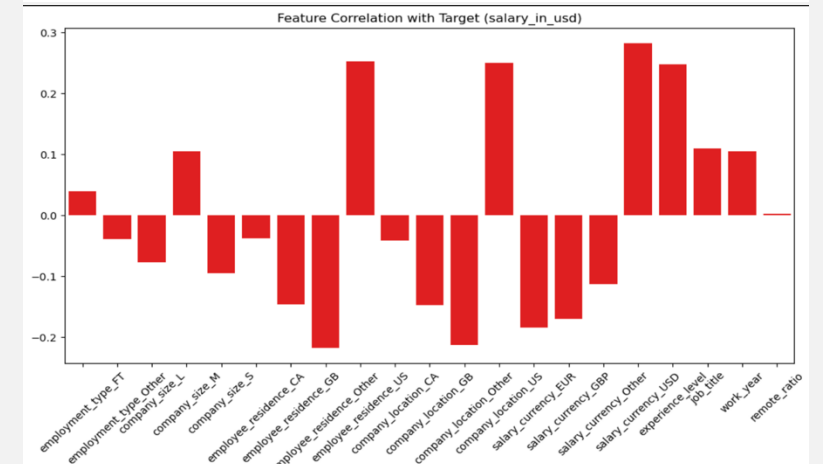
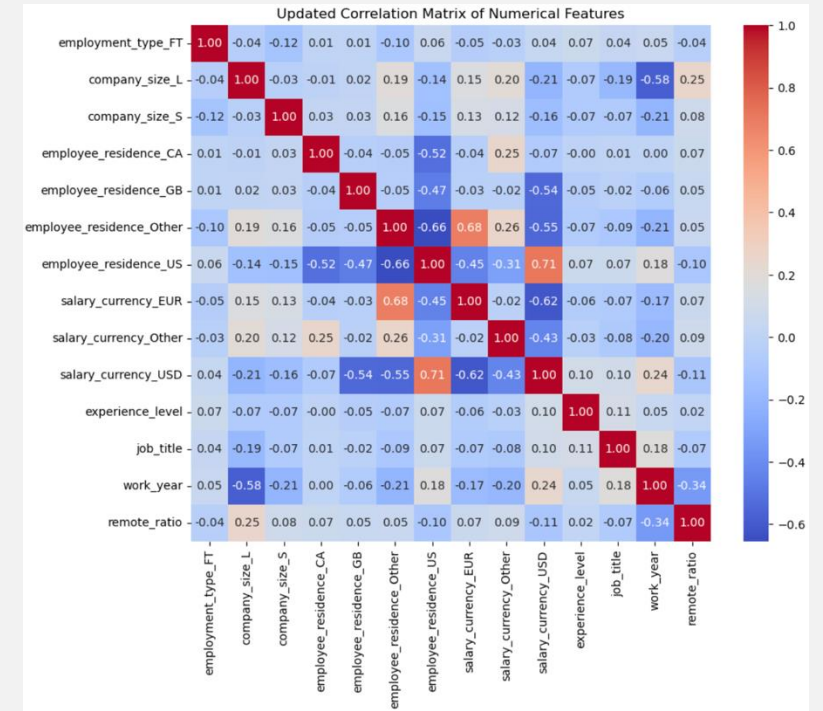
2.Scaling Numerical Features:

1. StandardScaler was applied to **work_year** and **remote_ratio**.
2. These features have different ranges, and scaling ensures the model treats them equally during training.



Data Preprocessing – Pearson Correlation

- **Correlation Matrix for Numerical Features:**
 - A heatmap was generated to visualize correlations between numerical features.
 - This helps identify relationships between independent variables.
- **Correlation with Target (salary_in_usd):**
 - A bar plot shows how each feature correlates with the target.
 - Positive correlations indicate features that increase with salary, while negative values suggest the opposite.
- **Threshold for High Correlation:**
 - Pairs of features with a correlation higher than **0.85** were flagged.
- **Results:**
 - Features such as **salary_currency_USD** and **employee_residence_US** show the strongest positive correlation with salary_in_usd.
 - No features had extreme correlations above the threshold, indicating no immediate need to drop significant columns.
 - This analysis ensures model stability and avoids overfitting by reducing redundant data.



Data Processing

Overall processing

Linear Regression

Random Forest

XGBoost

XAI

➡ Split dataset in $\leq 300K$ and $>300K$

➡ Recursive Feature Elimination

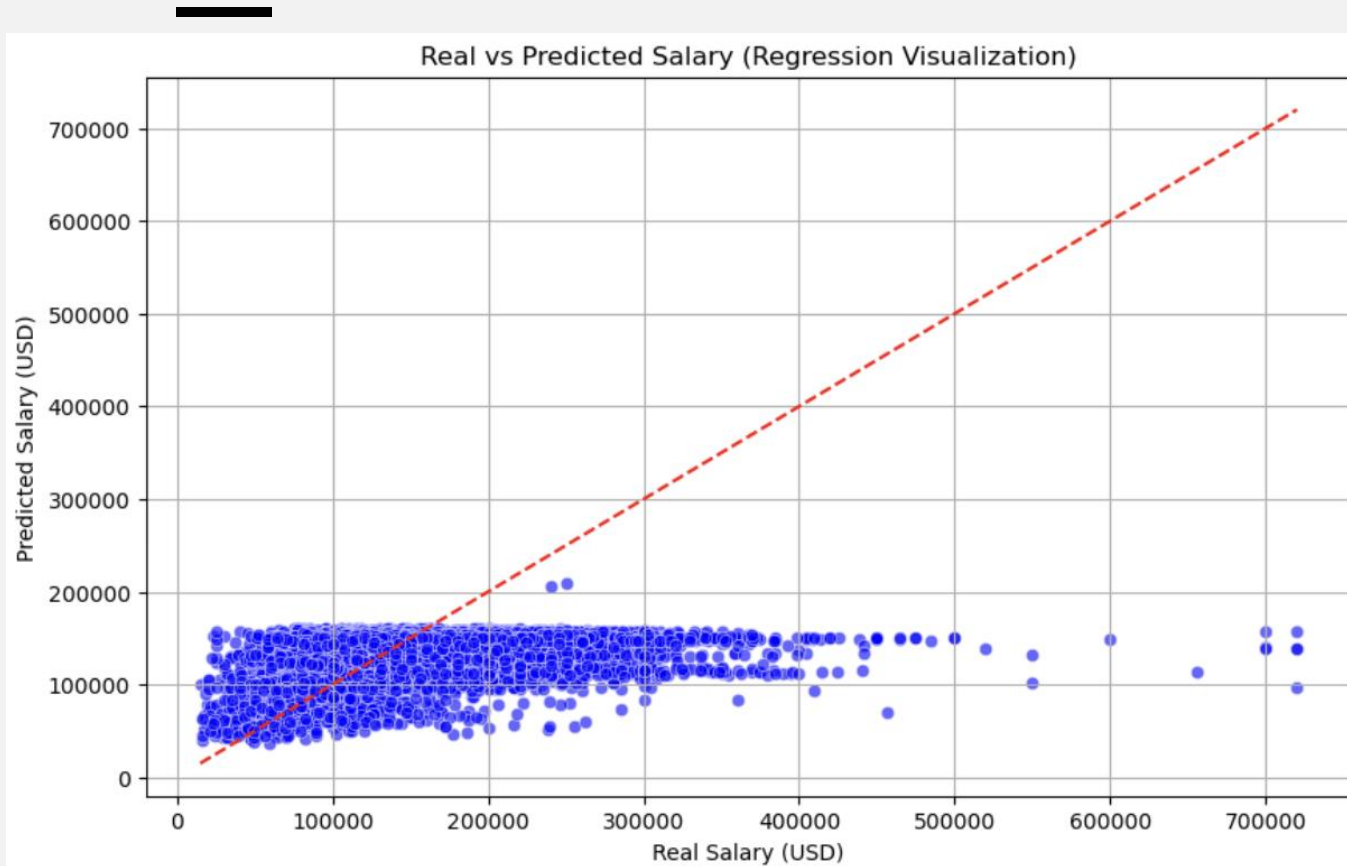
➡ RandomSearch CV

➡ RandomSearch CV

➡ Understanding Low Model Performance by Job Title

➡ Split dataset in $\leq 300K$ and $>300K$

Data Processing – Linear Regression



Results Interpretation:

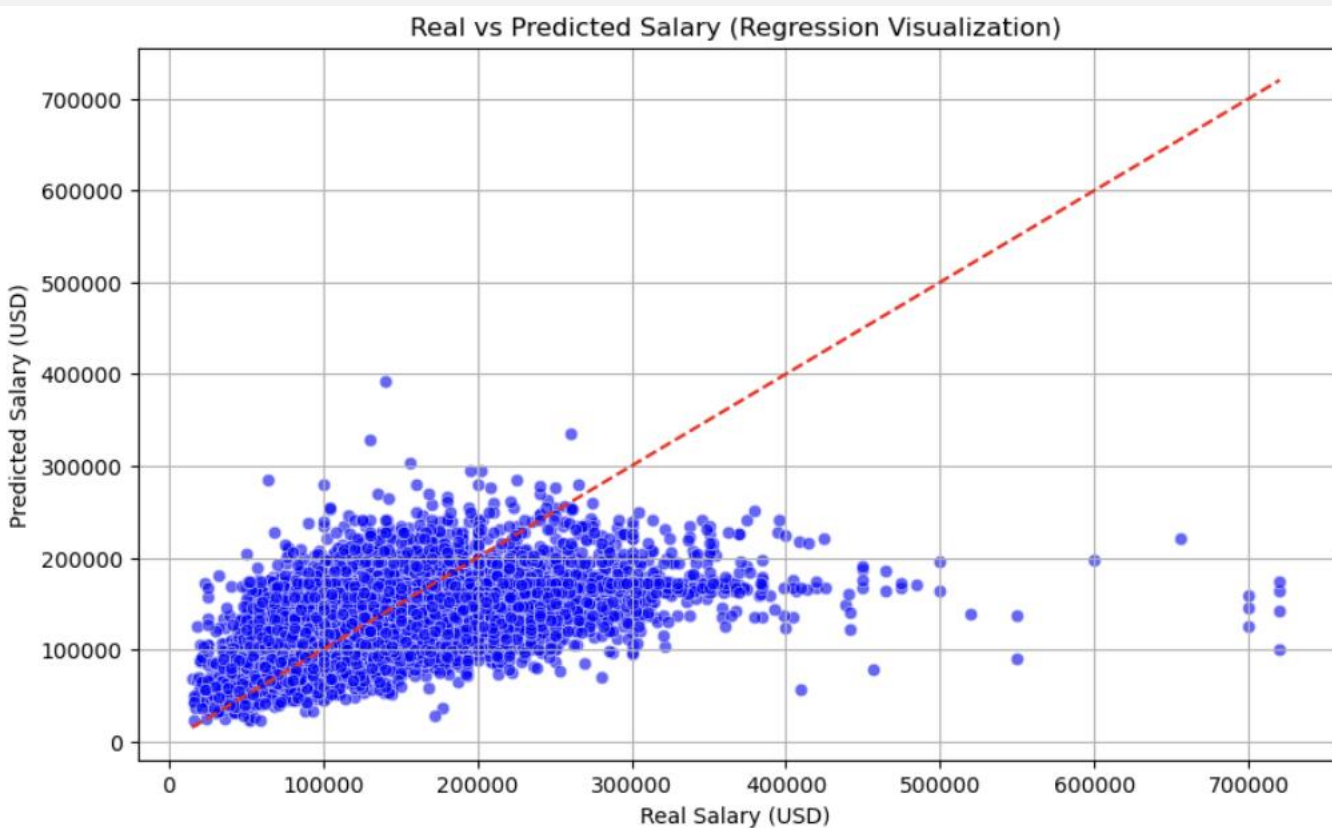
•Visual Analysis (Scatter Plot):

- The red dashed line represents the **ideal prediction ($y = x$)**.
- Most predictions cluster below the line, indicating the model tends to **underestimate higher salaries**.
- There are a few extreme outliers (salaries > 300k) poorly predicted by the model.

•Performance Metrics (Bottom Panel):

- **MAE:** 44,758 USD (average error in salary prediction).
- **RMSE:** 61,631 USD (penalizes larger errors).
- **R^2 :** 0.11 (the model explains only **11% of variance**, indicating low performance).

Data Processing – Random Forest



1. General Trend:

1. The model captures some degree of salary variation, but not with high precision.

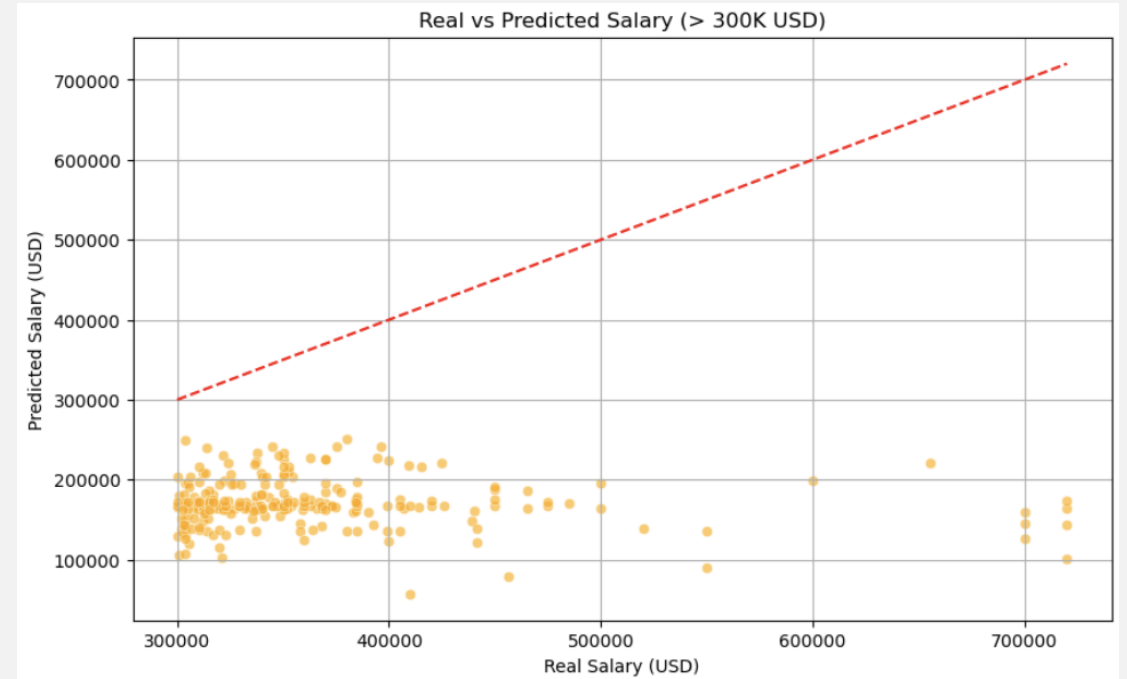
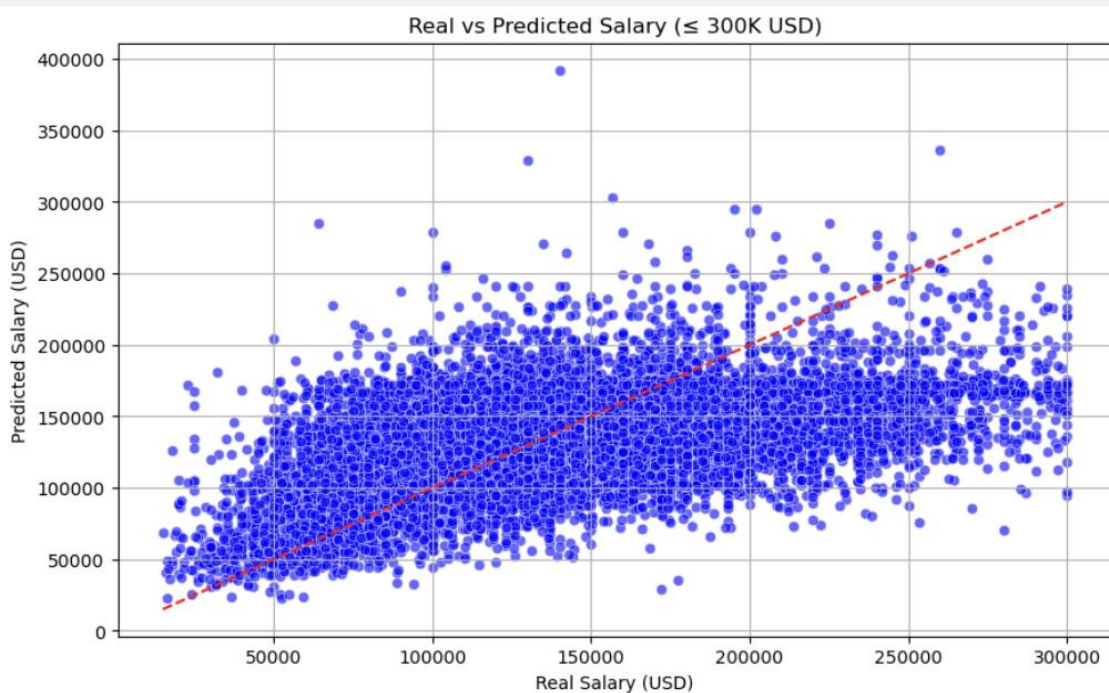
2. Deviations and Outliers:

1. For salaries above approximately \$300,000, predictions tend to flatten, suggesting **underestimation** of high salaries.
2. Outliers are present, particularly at higher salary ranges, where the model struggles to make accurate predictions.

3. Evaluation Metrics:

1. **Mean Absolute Error (MAE):** 43,018 USD
2. **Root Mean Squared Error (RMSE):** 57,973 USD
3. **R² (R-Squared):** 0.21 – Indicates that only **21% of the variance** in salaries is explained by the model. This is relatively low, showing room for improvement.

Data Processing – Random Forest split value



Results:

•Salaries $\leq$ \$300K:

- Model shows reasonable predictive capability with an R^2 of **0.22**.
- Errors are distributed relatively consistently, but some underestimation is observed at higher salary levels.

•Salaries $>$ \$300K:

- Model struggles significantly for high salaries, with an R^2 of **-6.04**.
- This suggests the need for more training data or different modeling approaches for extreme salaries.

Data Processing – Feature Selection

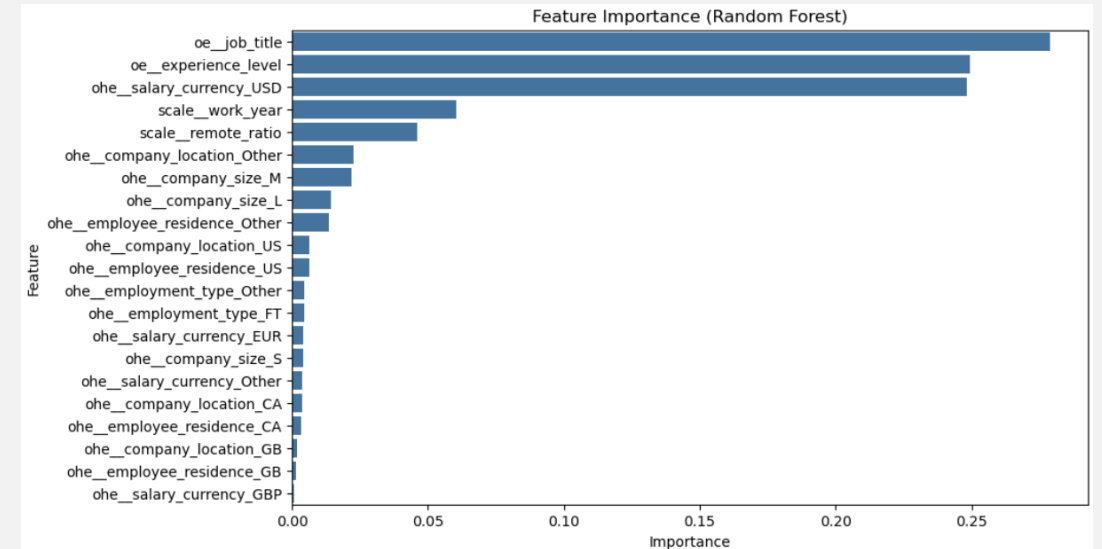
1. Random Forest Feature Importance

•How it Works:

- Random Forest provides a built-in feature importance mechanism by calculating how much each **feature decreases the impurity (Gini or variance)**.

•Results:

- The most important features were:
 - oe_job_title - oe_salary_currency_USD-oe_experience_level**
- Other factors, such as **company size** and **location**, had lower importance.



2. Permutation Importance

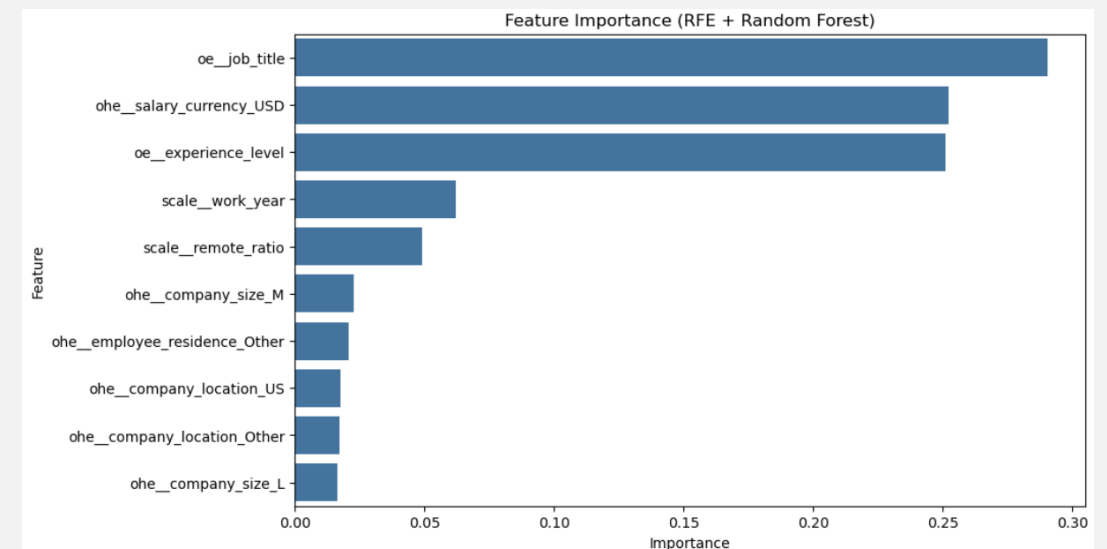
3. Recursive Feature Elimination (RFE)

•How it Works:

- RFE recursively **removes** the **least important features** by training the model **iteratively**. It continues until the desired number of features is selected.

•Results:

- The top 10 features selected through RFE align with the results of Feature Importance and Permutation Importance.
- This method reinforced the significance of **job title**, **salary currency**, and **experience level**.



Data Processing – RandomSearch CV

Hyperparameters Explained:

1.n_estimators – Number of trees in the forest.

1. Values tested: **[100, 200, 300, 400]**

2. **Best Value: 300**

2.max_depth – Maximum depth of the individual decision trees. (Prevent overfitting)

1. Values tested: **[10, 20, 30, 50, None]**

2. **Best Value: 30**

3.min_samples_split – Minimum number of samples required to split an internal node.

1. Values tested: **[2, 5, 10]**

2. **Best Value: 2** (aggressive splits to prevent underfitting)

4.min_samples_leaf – Minimum number of samples required to be in a leaf node.

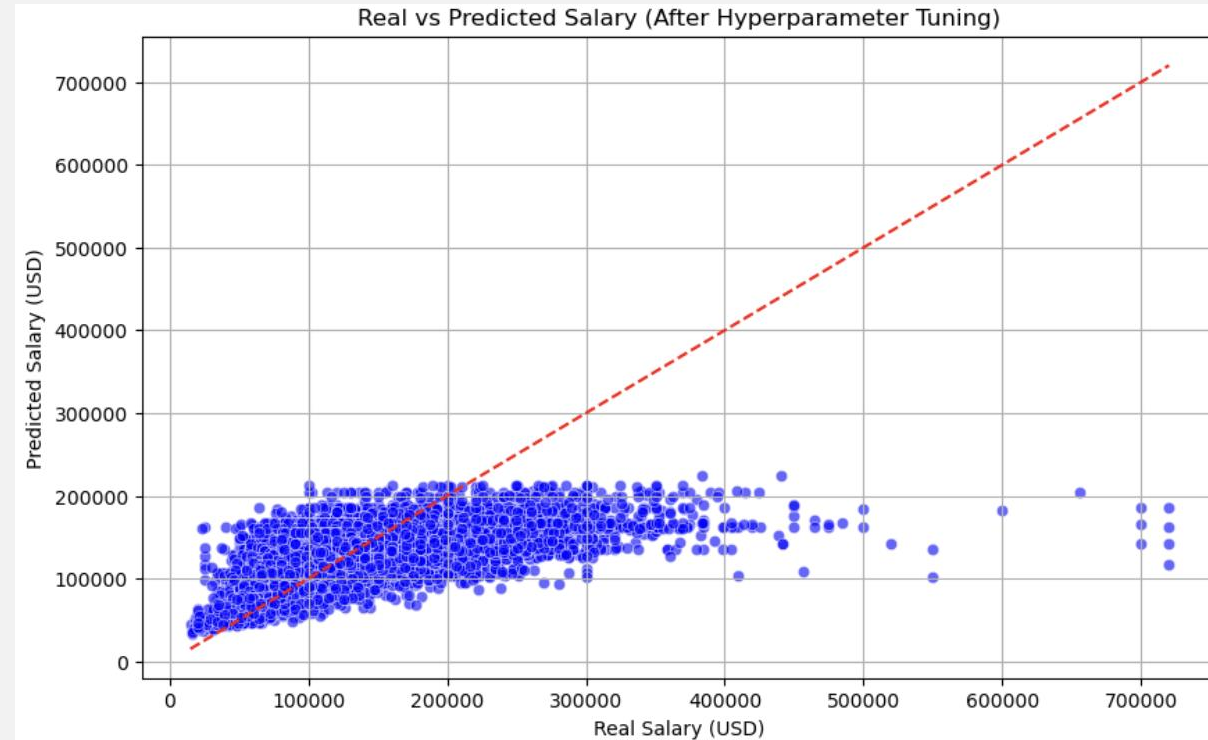
1. Values tested: **[1, 2, 4]**

2. **Best Value: 2**

5.max_features – Number of features to consider for the best split.

1. Values tested: **[sqrt, log2]**

2. **Best Value: sqrt** (uses the square root of the total features, balancing performance and speed)



Results After Tuning:

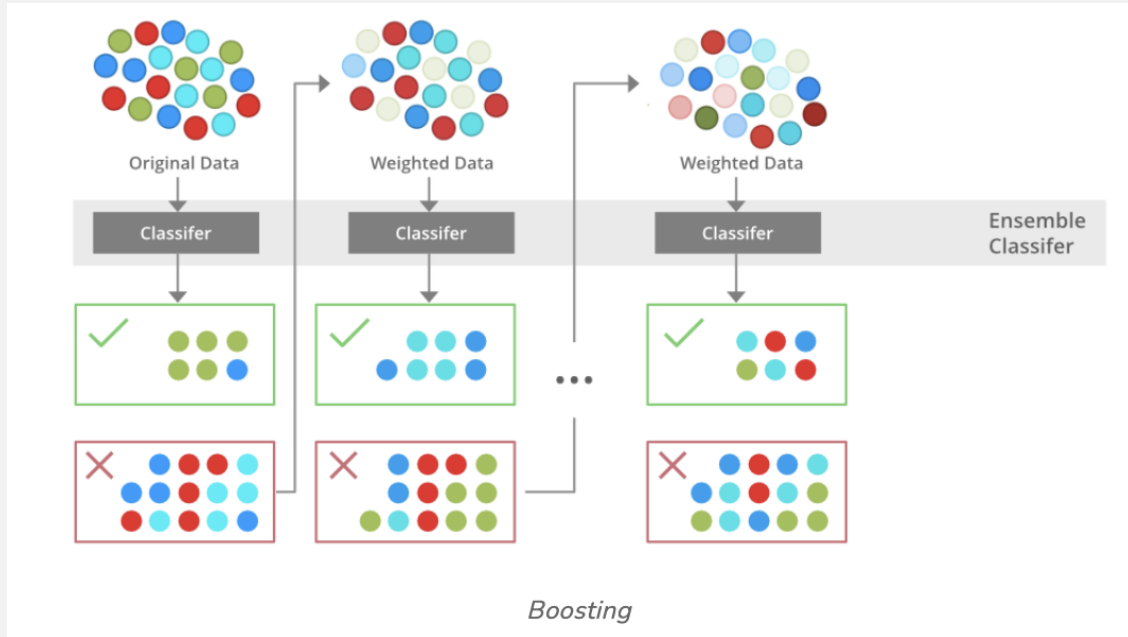
•MAE: 39,064 USD

•RMSE: 54,749 USD

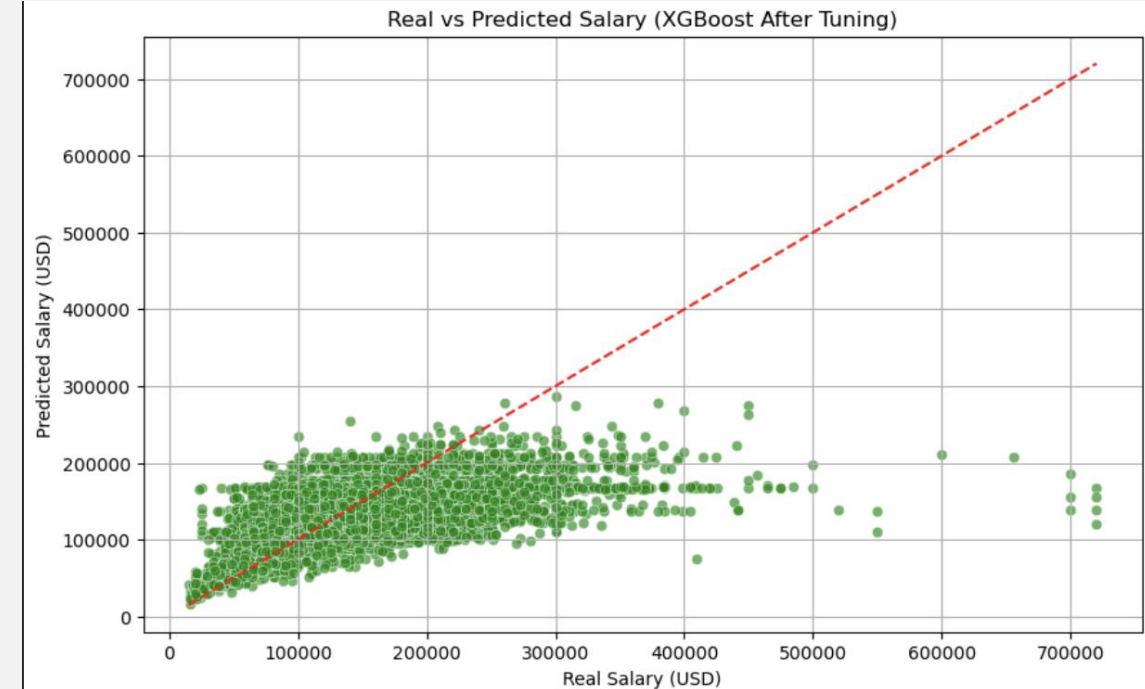
•R²: 0.2996

Data Processing – XGBoost RandomSearch CV

[1] Reference



- 1.First Classifier:** Trained on the original data. Misclassified samples are assigned **higher weights**.
- 2.Second Classifier:** Focuses on **correcting the errors** from the first classifier. Misclassified points continue to receive more weight.
- 3.Iterative Process:** This continues until a **strong ensemble classifier is created** by aggregating all weak learners.
- 4.Final Model:** The ensemble classifier makes the **final decision**, **combining** the outputs of all classifiers.



Results After Tuning:

- MAE: 39,551 USD
- RMSE: 54,630 USD
- R^2 : 0.3026

Data Processing – XGBoost split value



The model was divided into two segments:

1. Salary $\leq 300k$:

1. XGBRegressor **without** hyperparameter tuning.
2. Faster training and evaluation.

2. Salary $> 300k$:

1. XGBRegressor **with** hyperparameter tuning using RandomizedSearchCV.
2. Aimed at addressing the higher variance in this segment.

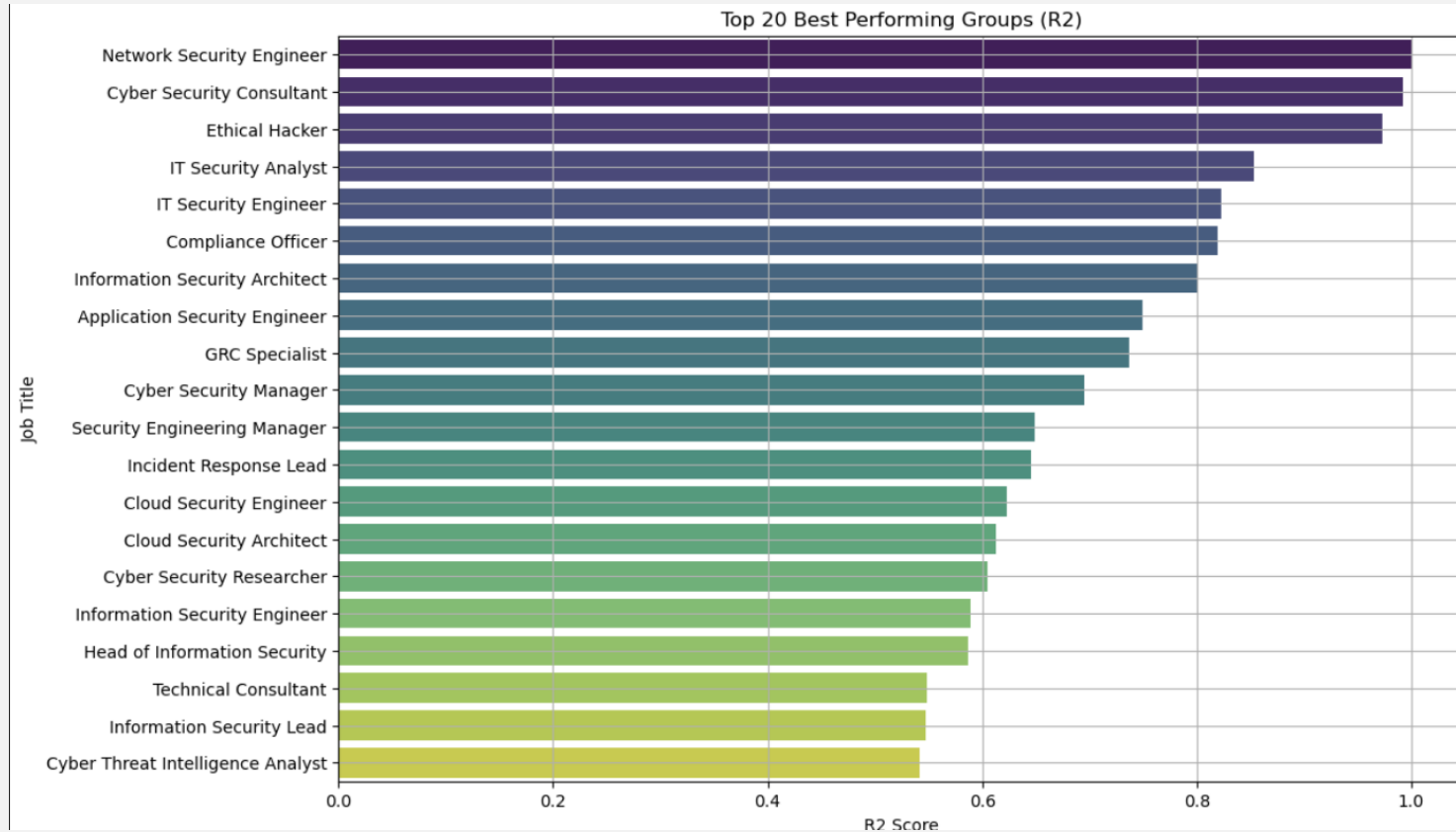
• Lower Salary Range ($\leq 300k$):

- **MAE:** 34,043 USD
- **RMSE:** 43,651 USD
- **R^2 :** 0.35
- The model performs better for this segment, achieving higher R^2 and lower errors.

• Higher Salary Range ($> 300k$):

- **MAE:** 34,931 USD
- **RMSE:** 54,747 USD
- **R^2 :** 0.41
- Predictions for higher salaries show slightly higher errors and lower R^2 , indicating challenges in modeling high-variance data.

Understanding Low Model Performance by Job Title



Motivation

To explain why these results were low and what caused the poor performance.

I identified that the variable **job_title** was the most significant feature in predicting salaries.

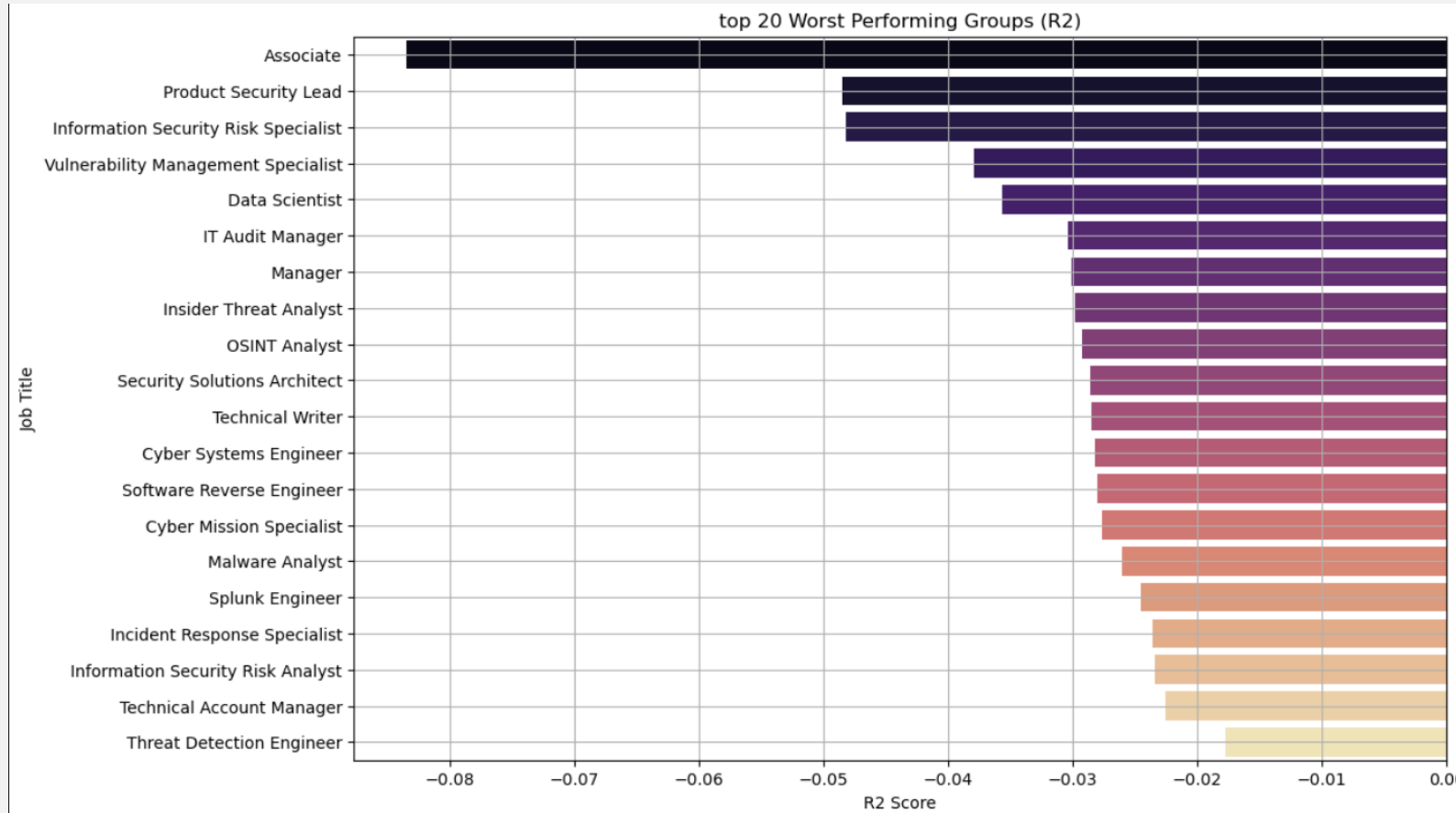
Based on this finding, I decided to **split** the dataset into **groups by job_title** to analyze whether individual job titles had distinct characteristics that could explain the variability in model performance.

Result

Job Titles with High R^2 :

1. Groups like "Network Security Engineer" and "Cyber Security Consultant" exhibited distinct and consistent patterns in features and salaries, The data for these groups showed **less variability** and **better alignment** with the features.

Understanding Low Model Performance by Job Title

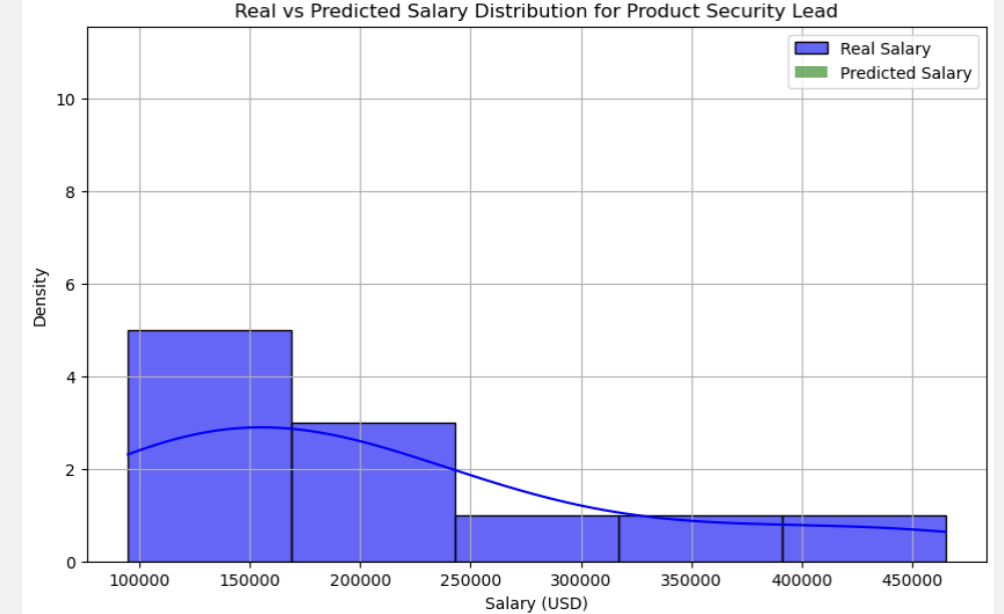
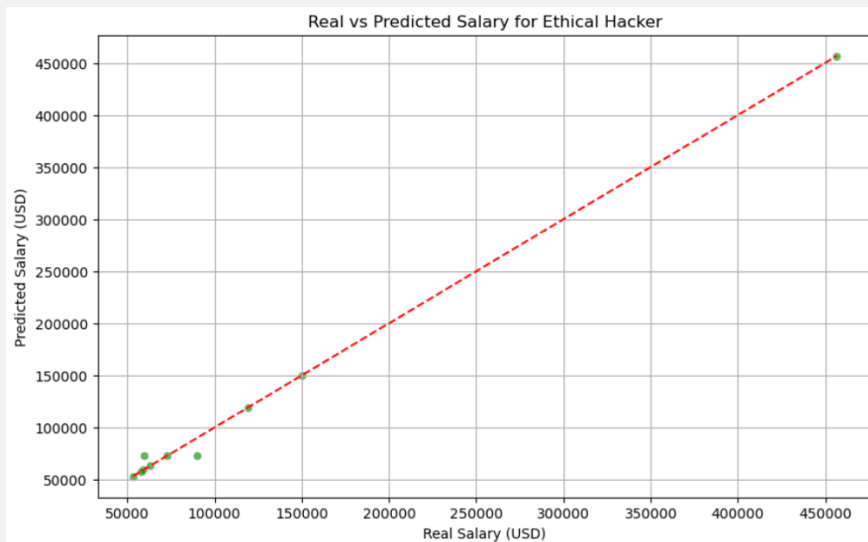
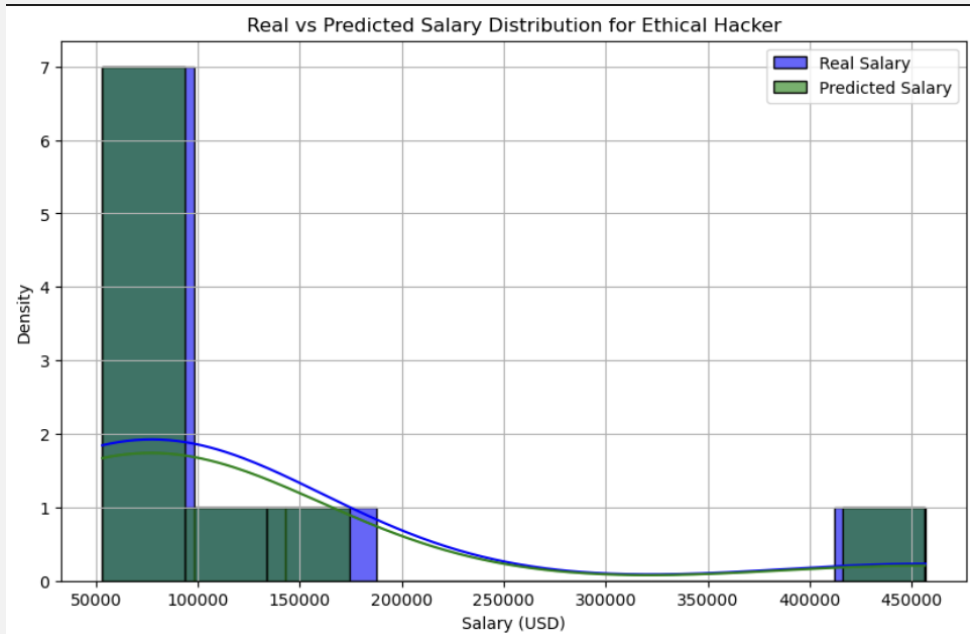


I observed that certain job titles had **negative R^2 scores**, indicating that the model performed worse than a simple baseline mean prediction for those groups.

Job Titles with Negative R^2 :

- Titles such as "Associate" and "Product Security Lead" showed particularly poor performance. These groups often had:
 - High salary variability**, making predictions more challenging.
 - Uneven or uninformative feature distributions**, limiting the model's ability to learn meaningful patterns.
- For some titles, the small number of observations amplified the effect of outliers.

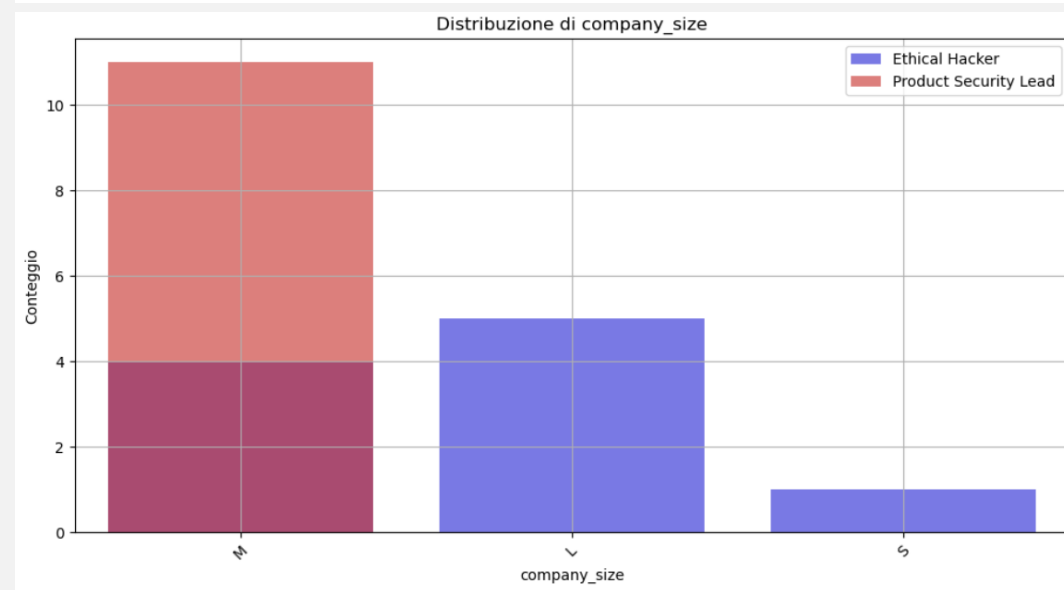
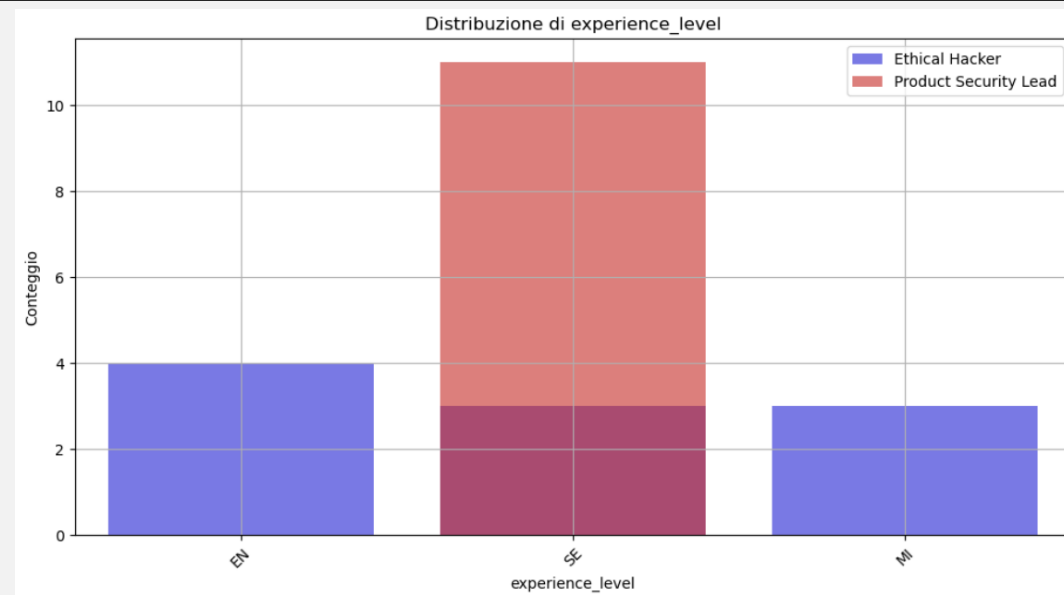
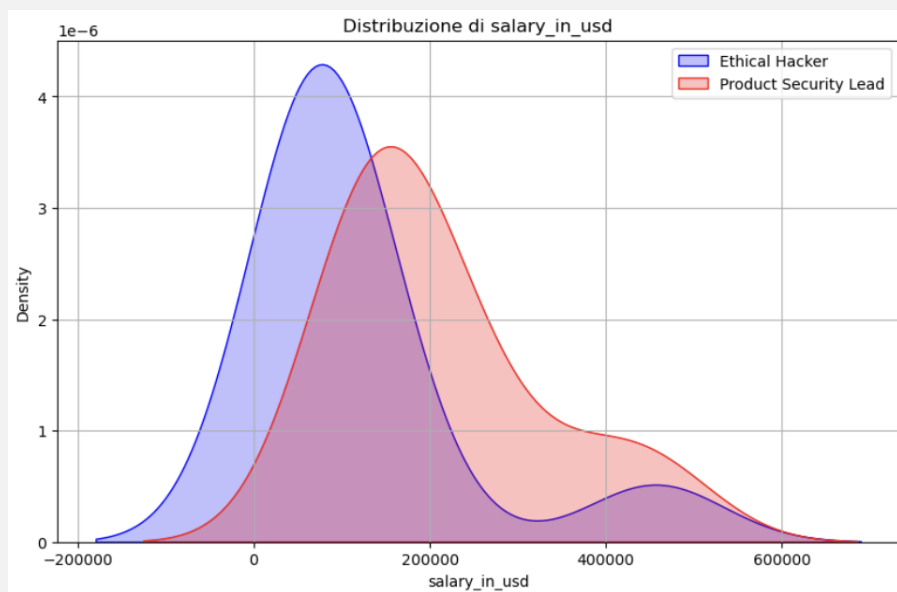
Understanding Low Model Performance by Job Title



Why these results?

To understand why the model performs well for some job titles and poorly for others, I selected two contrasting examples:

- **Ethical Hacker (R^2 high)**: This group represents a job title where the model performs well.
- **Product Security Lead (R^2 low)**: This group represents a job title where the model struggles to predict accurately.



Why these results?

Key Observations

1. Group Size:

1. PSL groups have small sample sizes: **11 for Product Security Lead**.

2. Salary Distribution:

1. **Ethical Hacker**: Salaries are concentrated within a narrower range (~50k to ~150k USD), creating a predictable pattern for the model.
2. **Product Security Lead**: Salaries are spread over a much wider range (up to 600k USD), introducing variability that the model struggles to explain.

3. Feature Distribution:

1. Experience Level:

1. **Ethical Hacker**: Balanced distribution across levels (EN, SE, MI).
2. **Product Security Lead**: Highly skewed toward senior-level positions (SE), reducing feature variability.

2. Company Size:

1. **Ethical Hacker**: Observations are distributed across small, medium, and large companies.
2. **Product Security Lead**: Dominated by medium-sized companies, limiting the information available for predictions.

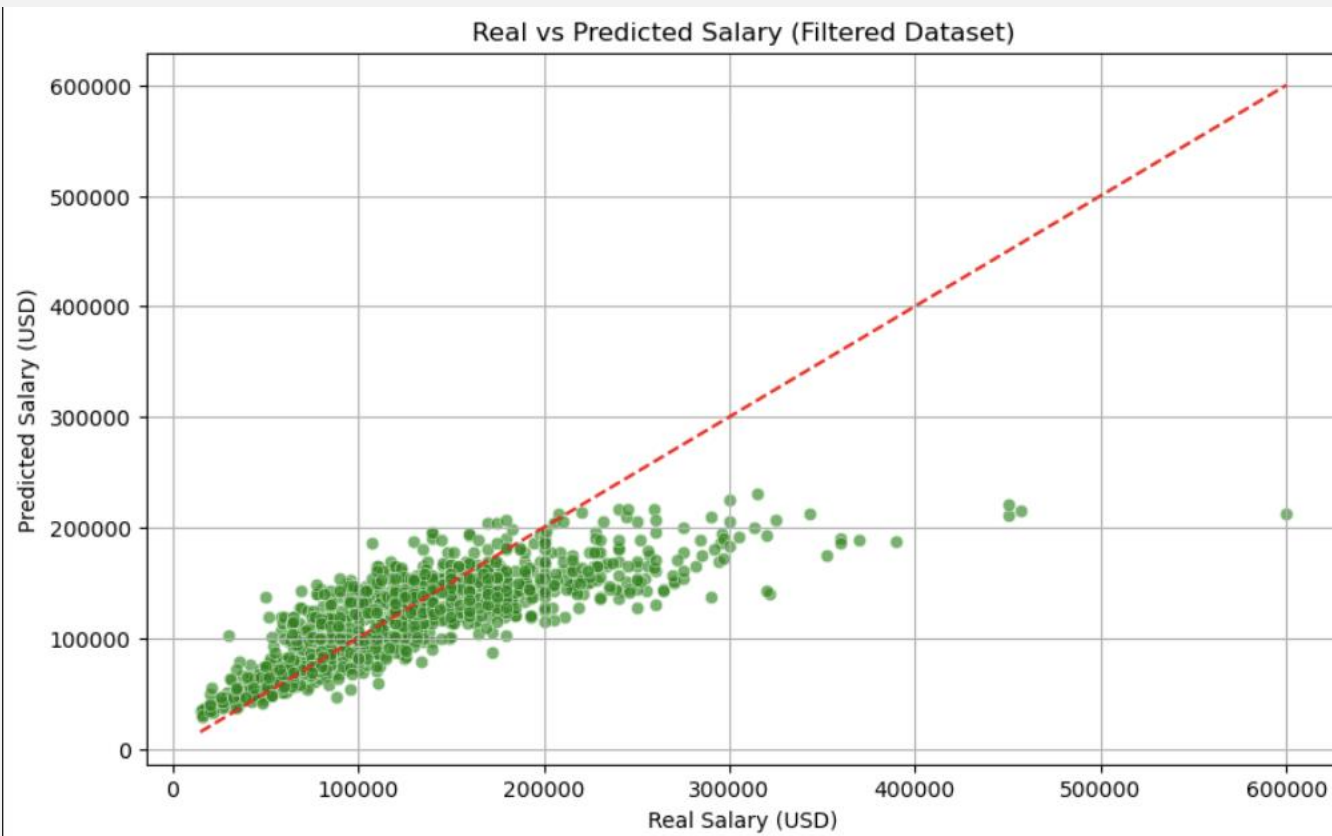
4. Features with Zero Variance:

1. For **Product Security Lead**, certain features such as:
 1. **work_year**: All observations have the same value, resulting in zero variance.
 2. **remote_ratio**: Similarly, this feature has no variability within the group.
2. Features with zero variance do not contribute to the model and reduce its ability to capture meaningful relationships, further lowering predictive performance.

The feature 'remote_ratio' for Product Security Lead have variance equal to zero. KDE not applicable.

The feature 'work_year' for Product Security Lead have variance equal to zero. KDE not applicable.

Understanding Low Model Performance by Job Title



```
Model Evaluation (Filtered Dataset):  
{ 'MAE': 28254.244290246603, 'RMSE': 41726.7464529429, 'R2': 0.5973524850304381 }
```

But,

Dimension of the dataset after filtering: (1214, 10)

1. Identification of Low-Performing Roles:

1. Job titles with an R^2 below 0.5 were identified and excluded from the dataset.
2. This step removed groups with significant variability or poor feature distributions that negatively impacted the model.

2. Filtered Dataset:

1. This filtering ensures the model focuses on data where it is likely to perform better.

3. Re-Training the Model:

1. The filtered dataset was used to re-train the model with **RandomizedSearchCV** for hyperparameter optimization.
2. A **5-fold cross-validation** was applied to ensure robustness in the evaluation.

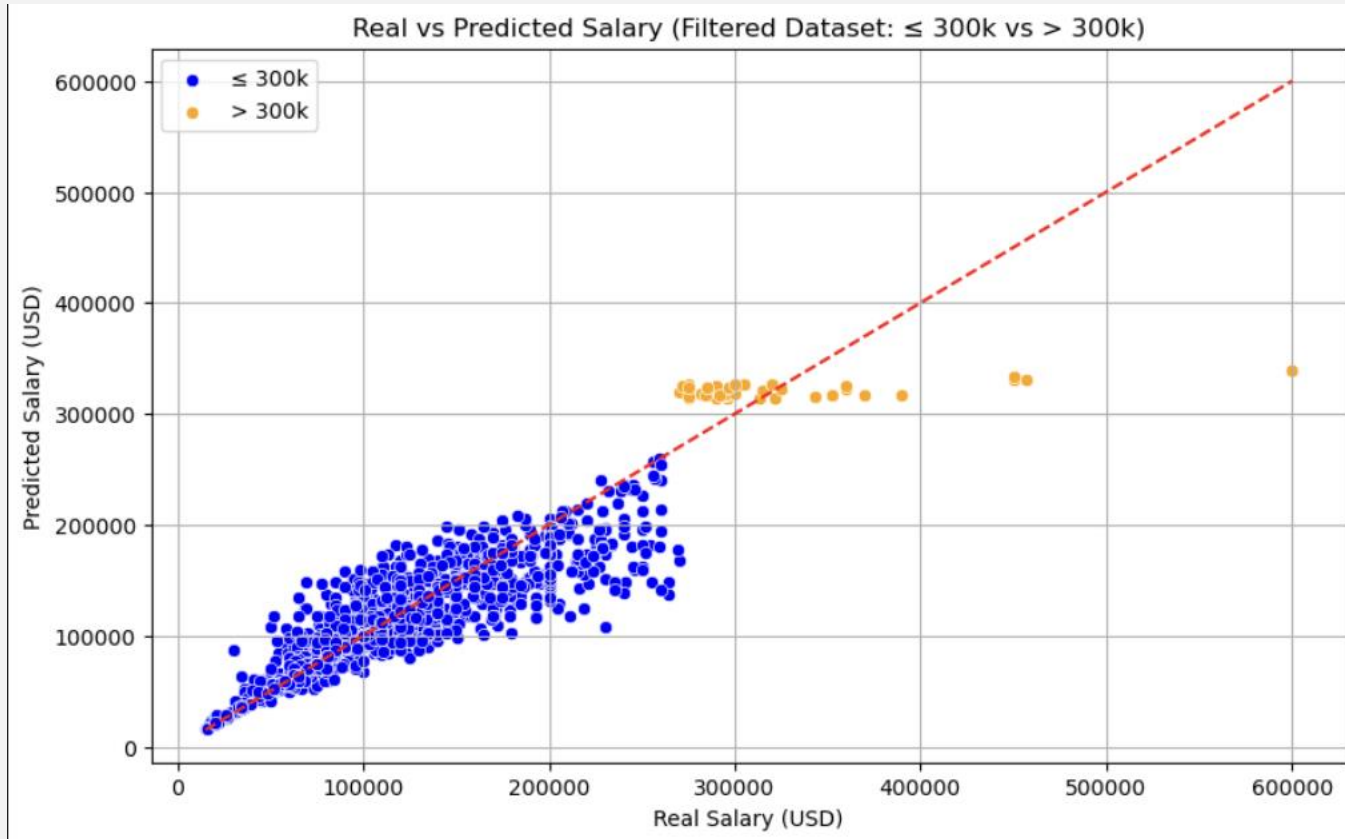
4. Evaluation on the Filtered Dataset:

1. A scatter plot of real vs. predicted salaries illustrates the model's improved performance.

Key Insights

- The **filtered dataset size (1,214 observations)** limits the model's ability to generalize, especially for the high-salary group, which comprises fewer data points.

Understanding Low Model Performance by Job Title



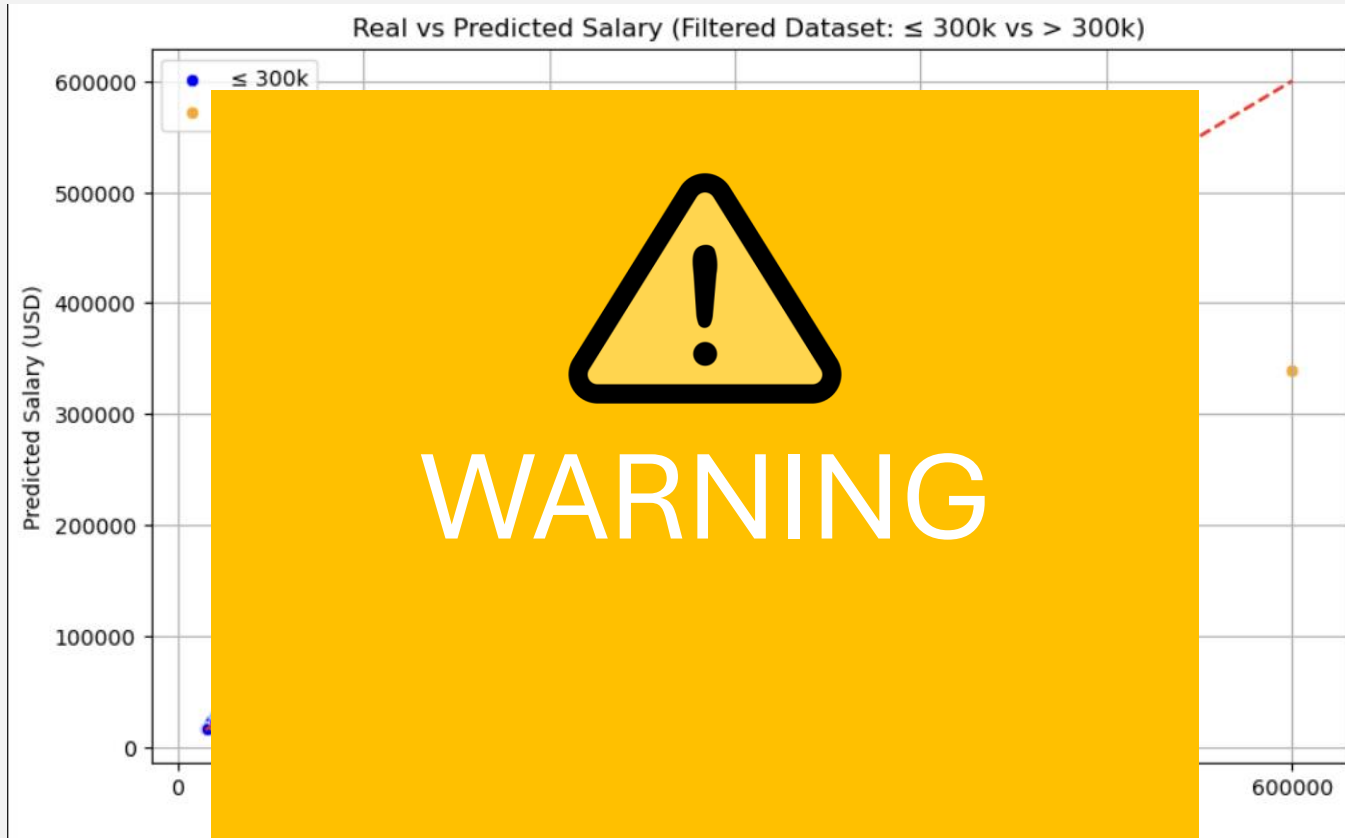
Model Evaluation (Filtered Dataset: $\leq 300k$):

```
{'MAE': 18879.118596338674, 'RMSE': 27689.50439446667, 'R2': 0.7462225367281938}
```

Model Evaluation (Filtered Dataset: $> 300k$):

```
{'MAE': 42679.076013513535, 'RMSE': 60389.90303512719, 'R2': 0.1832658170005026}
```

Understanding Low Model Performance by Job Title



Model Evaluation (Filtered Dataset: $\leq 300k$):

`{'MAE': 18879.118596338674, 'RMSE': 27689.50439446667, 'R2': 0.7462225367281938}`

Model Evaluation (Filtered Dataset: $> 300k$):

`{'MAE': 42679.076013513535, 'RMSE': 60389.90303512719, 'R2': 0.1832658170005026}`

Understanding Low Model Performance by Job Title



WARNING

After conducting extensive analyses and experiments with the dataset, it has become clear that **the dataset is not fully suitable for achieving consistently reliable and accurate results.**

```
Cross-Validation R2 (≤ 300k): 0.32
Cross-Validation R2 (> 300k): 0.00
Training R2 (≤ 300k): 0.75
Training R2 (> 300k): 0.18
```

1. Cross-Validation vs Training R² for ≤ 300k:

1. **Cross-Validation R²: 0.32**
2. **Training R²: 0.75**
3. The large gap between these values suggests **overfitting**:
 1. The model performs significantly better on the training data but fails to generalize to unseen data during cross-validation.
 2. This indicates the model has learned patterns specific to the training set (potentially noise) but cannot apply these patterns to new data.

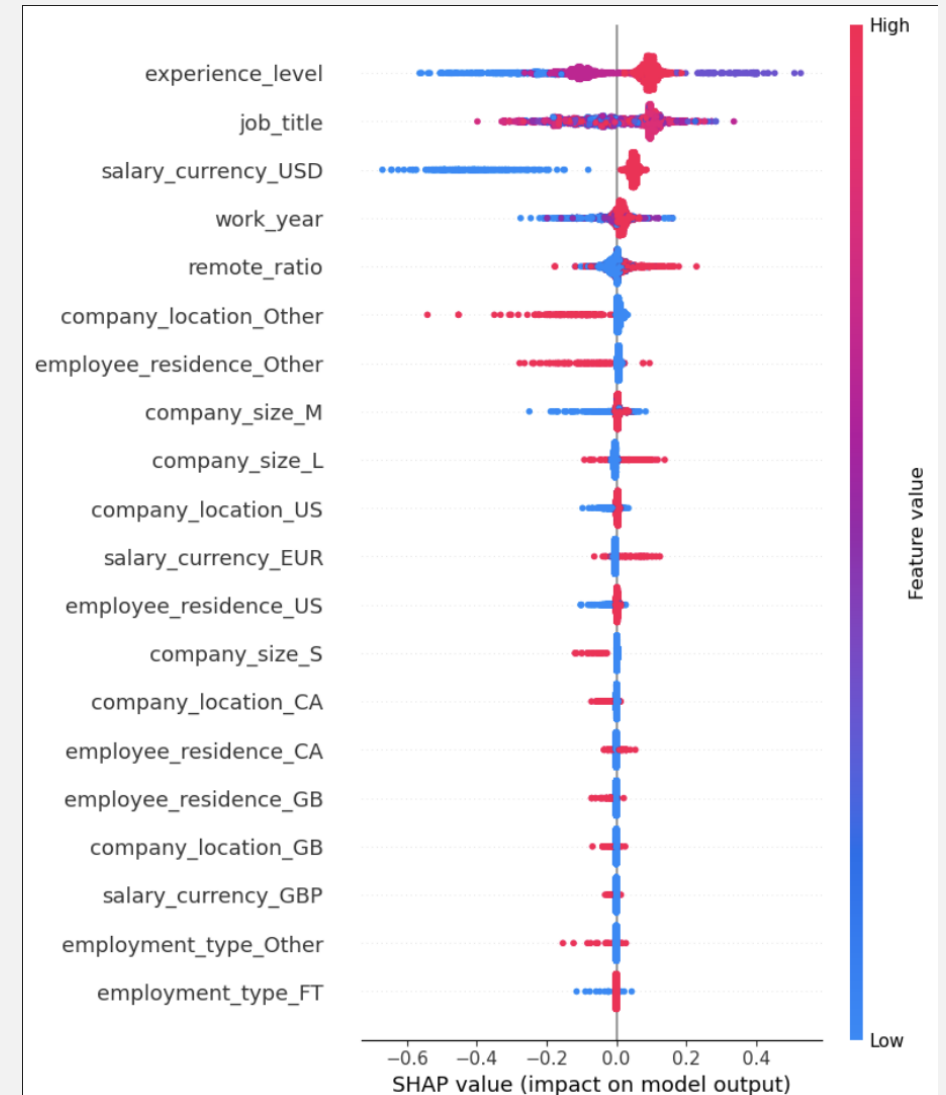
2. Cross-Validation vs Training R² for > 300k:

1. **Cross-Validation R²: 0.00**
2. **Training R²: 0.18**
3. In this range, the model struggles to learn meaningful patterns even on the training data, leading to poor generalization:
 1. The low R² values reflect the **limited predictive power of the model** for this segment of the dataset.
 2. This is likely due to the small sample size or high variability in salaries above 300k.

Data Processing – XAI

•SHAP:

- **Experience Level, Job Title, and Currency (USD)** are the most influential features and explainable of the model.
- Higher **remote ratio** negatively impacts predictions.



Result Summary

	MAE	RMSE	R ²
Linear Regression	44,758	61,631	0.11
Random Forest	43,018	57,973	0.21
RFE	43,050	57.005	0.21
RF with RandomSearch CV	39,064	54,749	0.29
XGBoost with RandomSearch CV	39,551	54,630	0.3026

XGBoost	split	MAE	RMSE	R ²
without	≤300K	34,043	43,651	0.35
with	>300K	34,931	54,747	0.41

Result Summary – T-test

Approach:

1. Conducted 5-fold cross-validation for each model.
2. Collected R^2 scores for each fold as the performance metric.
3. Applied **paired t-test** to compare the mean R^2 scores between models.

Comparison	P-value	Significant
Linear Regression vs Random Forest	0.0001	Yes
Linear Regression vs XGBoost	0.0002	Yes
Random Forest vs XGBoost	0.0024	Yes

Conclusion:

- **XGBoost** is the top-performing model, with statistically significant improvements over both **Random Forest** and **Linear Regression**.
- **Random Forest** also significantly outperforms **Linear Regression**, making it a strong alternative to XGBoost.

Model Performance Comparison with Baseline and Next step

```
RMSE score for train 47,002 USD/year, and for test 52,431 USD/year  
CPU times: user 2.88 s, sys: 460 ms, total: 3.34 s  
Wall time: 1.4 s
```

```
RMSE baseline score for train 64,600 USD/year, and for test 65,261 USD/year
```

[2] Reference

Conclusion:

- While the RMSE values may seem high, they are **reasonable** given the nature of the dataset and the task's complexity.
- The model is **more effective** than the baseline, demonstrating a better generalization to unseen data.

Future improvements could involve:

- **Collecting additional features** or external datasets.

Thank you for the attention!

- Reference:

[1]: <https://www.geeksforgeeks.org/xgboost/>

[2]: <https://www.kaggle.com/code/dima806/information-security-salary-autoviz-catboost-shap>

